

Roll No.

EVEN END SEMESTER EXAMINATION - 2025

Name of the Course: Propulsion 1

Semester: IV

Name of the Paper: Propulsion 1

Paper Code: **TAS 403**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Differentiate between Turbojet engine and Turbofan Engine used in Aircraft Propulsion. Write down the advantage and disadvantage of each.	
(b)	What do you mean by Bryton Cycle? Why it is used in gas turbine power plant, why not other cycle? Derive the efficiency formula of Bryton Cycle in terms of pressure ratio.	CO 2
(c)	What is the significance of all the laws of Thermo-dynamics in Aircraft Propulsion? Write down Steady flow energy equation. Air at 280K and 100KPa is compressed to 600KPa and 400K. Mass flow rate of air is 0.02 Kg per second and heat loss during compression is 16KJ/Kg. Consider the change in Potential and kinetic energy is negligible in the process of compression, find the power input required for compression.	
Q2	(20 marks)	
(a)	What do you understand by range of aircraft? Derive the equation of range of aircraft i.e. $S = \frac{u}{g} * \frac{1}{TSFC} * \frac{L}{D} * \ln\left(\frac{m_1}{m_2}\right)$ with all the required steps. An aircraft cruises at a speed of 250 m/s. It has a TSFC of 0.0002 per sec, and a lift-to-drag ratio L/D=15. The aircraft's initial mass is 80,000 kg and final mass is 60,000 kg. Find the range of the aircraft.	CO4
(b)	The Boeing 747 aircraft is powered by four CF-6 turbofan engine manufactured by General Electric Each engine has following data:	

Thrust force	30.0 KN
Air mass flow rate	150 kg/s
Bypass ratio	5
Fuel mass flow rate	0.75 kg/sec
Operating Mach number	0.8
Altitude	10 km
Ambient Pressure	24.6 kPa
Ambient temperature	223.3 K
Calorific heat of fuel(Q _{cv})	48000 KJ/kg

If the thrust generated by fan is 75% of the total thrust, determine the following

- 1: Jet velocity of cold and hot gases
- 2: The specific thrust
- 3: The TSFC
- 4: Propulsive efficiency
- 5: Thermal efficiency
- 6: Overall efficiency

- (c) Discuss in detail about all the factors that affects the thrust force of Jet engine. Includes all the parameters like jet nozzle, airspeed, mass flow rate of air, altitude, ram effect.

Q3

(20 marks)

(a)

What do you understand by area velocity relation of Nozzle flow? Derive the equation of Area velocity Relation for the same up to final result $\frac{dA}{A} = (M^2 - 1) * \frac{du}{u}$. And with the reference of Area velocity relation discuss all the cases and converging and diverging duct act as nozzle and diffuser.

CO4

(b)	Differentiate between Polytropic efficiency and Isentropic efficiency terms. Derive the formula of Isentropic efficiency of Compressor and Turbine with Suitable t-s plot.	
(c)	A Ramjet is travelling at Mach 4 at an altitude of 4572m, the external static temperature is 258.4 K, and the external static pressure is 57.10 kPa. The heating value of the fuel is 48,000 KJ/kg. Air flows through engine at 45.35 kg/s. The burner exit temperature is 2000 K. Find the Thrust, fuel-air ratio and TSFC. The specific heat ratio is assumed to be 1.40	
Q4	(20 marks)	
(a)	A centrifugal Compressor has inlet eye diameter of 20 cm. It runs at speed of 21000 rpm and takes air at 110m/sec axial speed. Inlet stagnation condition are 300 K and 1.03 bar. Find the blade angle at inlet and the Mach number at the tip of the impeller eye.	CO5
(b)	Air at a temperature of 300K enters a ten stage axial flow compressor at the rate of 3kg/sec. the pressure ratio is 7 and the isentropic efficiency is 90%, the compression process being adiabatic. The compressor has symmetrical blades. The axial velocity of 110 m/s is uniform across the stage and the mean blade speed of each stage is 180 m/sec. Determine the direction of the air at entry to and exit from the rotor and the stator blades and also power given to air. Assume $C_p=1.005$ KJ/kgK and Gamma is 1.4	
(c)	A multistage gas turbine is to be designed with impulse stages, and is to operate with an inlet pressure and temperature of 6 bar and 1123K and an outlet pressure of 1 bar. The isentropic efficiency of the turbine is 88%. All the stages to have a nozzle outlet angles of 65 degree and equal outlet and inlet angles. Mean blade speed of 310m/s and equal inlet and outlet gas velocities. Assume $c_p=1150$ J/kgK and $\gamma=1.33$ for the optimum blade speed condition.	
Q5	(20 marks)	
(a)	What do you understand by turbine cooling and what are the various methods for turbine cooling ? explain in detail	CO 3
(b)	Define the term Cascade. Explain in detail about turbine Cascade and Compressor Cascade.	
(c)	Explain in detail about various types of cascade variable with their effects.	

